Week 48: Inside and outside covers

Adult guide built from the problem files · Unpiloted. Physical pretests unperformed.

This week's adult guide is week-48-facilitator.pdf, whose source is editable/facilitator-src/content.json with route-note.json. These problem files hold each problem's one-sentence idea, not its answers; the answers, hints and the budget proof are in the guide.

Problems and routes

Problem	K–1	2–3	4–5	route
Inside and cover for the triangle	P1	P1	P1	first
Split two big tiles of the trapezoid cover	P2			return
Trap the diamond's area		P2	P2	first
The triangle with small squares	P3	P3	P3	first
The diamond with small squares	P4	P4	P4	K-1 return; 2-3 first; 4-5 first
The trapezoid on both grids		P5		return
Same markings, different areas	P5	P6		return
What the markings guarantee	P6	P7		return
Spend three splits on the trapezoid			P5	return
Can a split make a bound worse?			P6	return
The trapezoid's exact area			P7	return

First route. K–1 P1, P3; Grades 2–3 P1–P4; Grades 4–5 P1–P4. **Return visits.** K–1 P2, P4–P6; Grades 2–3 P5–P7; Grades 4–5 P5–P7.

Key, one entry per problem

Inside and cover for the triangle

K–1 P1 · Grades 2–3 P1 · Grades 4–5 P1 · first route

Idea. On the 4-by-4 grid the half-square triangle has 6 whole cells and a cover of 10, so its area lies between 6 and 10.

Split two big tiles of the trapezoid cover

K–1 P2 · return route

Idea. Splitting a partial big tile lets its outside small tiles go; on the trapezoid the corner tiles each free three, so two splits remove at most six small tiles.

Trap the diamond's area

Grades 2–3 P2 · Grades 4–5 P2 · first route

Idea. The diamond has 4 whole cells and a cover of 12, so its area is certainly more than 3 and less than 13.

The triangle with small squares

K–1 P3 · Grades 2–3 P3 · Grades 4–5 P3 · first route · wording differs by band

Idea. Small squares raise the triangle's inside count from 6 to 7 and lower its cover from 10 to 9 big-square units; refining never loosens either bound.

The diamond with small squares

K–1 P4 · Grades 2–3 P4 · Grades 4–5 P4 · K-1 return; 2-3 first; 4-5 first route · wording differs by band

Idea. On the small grid the diamond's bounds become 6 and 10, so the uncertain area falls from 8 to 4 big-square units; K-1 changes the shape without changing its big-square marking.

The trapezoid on both grids

Grades 2–3 P5 · return route

Idea. The trapezoid's bounds go from 4 and 16 on the big grid to 6 and 9 on the small grid, so its uncertain area falls from 12 to 3: splitting every square does not always halve the gap.

Same markings, different areas

K–1 P5 · Grades 2–3 P6 · return route · wording differs by band

Idea. With w whole and p partial squares, any area strictly between w and w + p is possible, so the same markings allow very different areas.

What the markings guarantee

K–1 P6 · Grades 2–3 P7 · return route · wording differs by band

Idea. The markings guarantee only $w < \text{area} < w + p$: K-1 enlarges a shape without a new whole square; 2-3 finds that 4 whole and 4 partial squares allow any area strictly between 4 and 8.

Spend three splits on the trapezoid

Grades 4–5 P5 · return route

Idea. A split removes at most its own cell's uncertainty, so the best three splits on the trapezoid are the cells whose children all become whole or outside.

Can a split make a bound worse?

Grades 4–5 P6 · return route

Idea. A whole cell's four children are whole and a cover cell's children can only be kept or dropped, so splitting never lowers the inside area or raises the cover.

The trapezoid's exact area

Grades 4–5 P7 · return route

Idea. The trapezoid is a 2-by-3 rectangle and two triangles of 3/4, area 15/2, which lies between the bounds 4 and 16 and between the bounds after three splits, 5.5 and 14.5.